SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

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COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO* (BL3, BL6)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **TOTAL MARKS: 25**

Annexure A
Industry Problem-Solving Task

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

1. SSL/TLS Overhead and Web Server Performance Optimization
A web server uses TLS 1.3 for secure communication. Each HTTP response carries 2000 bytes of payload. TLS adds 25 bytes (record header + authentication tag) and handshake overhead of 3 KB per session. The server handles 10,000 requests per second, with 1,000 new TLS handshakes per second.

Task:
A. Calculate the total data transmitted per second including TLS overhead.
B. Determine the percentage overhead due to TLS encryption.
C. Estimate bandwidth consumed due to handshake overhead per second.
D. Analyze the trade-off between security and latency, and propose optimization techniques (e.g., session resumption, HTTP/2, or QUIC).

ANSWER:

Given

- HTTP payload per response = **2000 bytes**
- TLS record overhead = **25 bytes**
- Requests per second = **10,000**
- New TLS handshakes per second = **1,000**
- Handshake overhead = **3 KB/session**

A. Total data transmitted per second

TLS adds 25 bytes to every response:

Total per response=2000+25=2025 bytes

For 10,000 requests/sec:

2025×10000=20,250,000 bytes/sec

Therefore:

20,250,000 bytes/sec

Approximately:

20.25 MB/s

In bits:

$20{,}250{,}000 \times 8 = 162{,}000{,}000$ bits/sec 162 Mbps

Answer: 20.25 MB/s or 162 Mbps for the HTTP responses including TLS record overhead.

B. Percentage overhead due to TLS encryption

TLS adds **25 bytes** to a 2000-byte payload.

$$\text{Overhead } \% = \frac{25}{2000} \times 100 = 1.25\%$$

Therefore:

1.25%

So, the TLS record overhead increases the response size by **1.25%**.

C. Bandwidth consumed by handshake overhead

Each TLS handshake adds **3 KB**.

Using:

$1 \text{ KB} = 1024$ bytes $3 \text{ KB} = 3 \times 1024 = 3072$ bytes

There are 1,000 new handshakes/sec:

$3072 \times 1000 = 3{,}072{,}000$ bytes/sec

Therefore:

3,072,000 bytes/sec

or approximately:

3.072 MB/s

In Mbps:

$3{,}072{,}000 \times 8 = 24{,}576{,}000$ bits/sec 24.576 Mbps

Total including handshake traffic

If both response overhead and handshake traffic are considered:

$20{,}250{,}000 + 3{,}072{,}000 = 23{,}322{,}000$ bytes/sec ≈ 23.32 MB/s or

≈ 186.58 Mbps

D. Security vs. Latency Trade-off

TLS provides important security benefits:

- **Confidentiality** – prevents attackers from reading data.
- **Integrity** – prevents undetected modification.
- **Authentication** – certificates help verify the server.
- **Protection against network attacks** – reduces risks such as eavesdropping and man-in-the-middle attacks.

However, TLS introduces some overhead.

Main performance issues

1. Handshake latency

A new TLS connection requires a handshake, which consumes CPU, bandwidth, and network round trips.

With **1,000 new handshakes/sec**, the server experiences significant handshake overhead.

2. Encryption overhead

Every response contains 25 bytes of TLS overhead. Here, this is relatively small:

1.25%

3. CPU overhead

Encryption/decryption and authentication require server processing, particularly when many new TLS sessions are created.

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2. PGP-Based Email Encryption Load in Enterprise System
An enterprise email server encrypts emails using PGP. Each email contains a 10 MB attachment. PGP introduces 800 bytes overhead for key encryption and signatures. The system processes 5,000 emails per day.

Task:
A. Calculate the total data transmitted per day including overhead.

B. Determine the percentage increase in storage due to PGP.
C. Analyze the computational overhead during encryption/decryption.
D. Recommend optimization strategies (e.g., compression, batching, or hardware acceleration).

ANSWER:

Given

- Attachment size = **10 MB/email**
- PGP overhead = **800 bytes/email**
- Emails processed = **5,000 emails/day**
- Assume **1 MB = 1,000,000 bytes**

A. Total Data Transmitted per Day

Size of one email including PGP overhead:

$10,000,000+800=10,000,800$ bytes

For 5,000 emails:

$10,000,800\times5,000=50,004,000,000$ bytes

Therefore:

50,004,000,000 bytes/day

or approximately:

50.004 GB/day

Without PGP overhead:

$10,000,000\times5,000=50,000,000,000$ bytes=50 GB

So, PGP adds:

4,000,000 bytes=4 MB/day

B. Percentage Increase in Storage

PGP overhead per email:

800 bytes

Percentage increase:

$\frac{800}{10,000,000} \times 100 = 0.008\%$

Therefore:

0.008%

The storage overhead caused by PGP is **very small** compared with the 10 MB attachment size.

C. Computational Overhead

PGP uses **hybrid encryption**, combining symmetric and asymmetric cryptography.

Encryption

1. A random symmetric session key is generated.
2. The 10 MB attachment is encrypted using the symmetric key.
3. The session key is encrypted using the recipient's public key.
4. A digital signature may be generated using the sender's private key.

Decryption

1. The recipient uses their private key to recover the session key.
2. The session key decrypts the 10 MB attachment.
3. The digital signature is verified.

The major computational cost for the large attachment comes from **symmetric encryption/decryption**, because the entire 10 MB file must be processed.

Asymmetric operations such as RSA/ECC are performed mainly on the **small session key and signatures**, rather than on the entire 10 MB attachment.

For 5,000 emails:

$10\text{ MB} \times 5000 = 50,000\text{ MB} = 50\text{ GB/day}$

Thus, the system must encrypt/decrypt approximately **50 GB of attachment data per day**.

D. Optimization Strategies

1. Compression

Compress attachments before encryption.

Original File
↓
Compression
↓
Encryption
↓
PGP Email

Compression can reduce:

- Storage requirements
- Network bandwidth
- Encryption workload

Important: Compression is generally more effective **before encryption**, because encrypted data has high entropy and usually does not compress well.

2. Batching

Group encryption operations where appropriate instead of repeatedly initializing expensive processing pipelines.

This can reduce per-message processing overhead, although each recipient's encryption and authentication requirements still need to be preserved.

3. Hardware Acceleration

Use CPUs with cryptographic acceleration or dedicated cryptographic hardware to improve symmetric encryption/decryption throughput.

4. Efficient Algorithms

Use modern symmetric algorithms such as **AES** for large attachments and efficient public-key algorithms such as **ECC** for key protection/signatures where supported.

5. Parallel Processing

Since emails are independent, multiple messages can be encrypted/decrypted concurrently using worker processes or threads.

5000 Emails
↓

Worker 1	Worker 2	Worker 3	Worker 4

↓
Faster Processing

Final Answers

Task	Result
A. Total data/day including PGP	**50.004 GB/day**
PGP overhead/day	**4 MB/day**
B. Storage increase	**0.008%**
C. Data encrypted/day	**50 GB/day**
D. Recommended optimization	**Compression, parallel processing, batching, hardware acceleration**

Conclusion

PGP adds only **0.008% storage overhead** to each 10 MB email, so its bandwidth/storage impact is negligible. The main enterprise challenge is **computational load from processing approximately 50 GB of attachments per day**. Compression before encryption, parallel processing, efficient cryptographic algorithms, and hardware acceleration can significantly improve system performance while maintaining PGP security.

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3. IPSec Transport vs Tunnel Mode Performance Comparison
A company compares IPSec transport mode and tunnel mode for securing internal communications. Original packet size = 1000 bytes payload + 20-byte header.
Transport mode adds 30 bytes, while tunnel mode adds 50 bytes overhead. Traffic volume = 20,000 packets per second.

Task:
A. Compute total packet size for both modes.
B. Calculate bandwidth usage for each mode per second.
C. Compare efficiency and security benefits of both modes.
D. Recommend the appropriate mode for internal vs external communication with justification.

ANSWER:

Given

- Payload = **1000 bytes**
- Original IP header = **20 bytes**
- Original packet size:

1000+20=1020 bytes

- Transport Mode overhead = **30 bytes**
- Tunnel Mode overhead = **50 bytes**

- Traffic = **20,000 packets/second**

A. Total Packet Size

1. Transport Mode

Transport mode keeps the original IP header and adds 30 bytes of IPSec overhead.

$1020+30=1050$ bytes Transport packet size=1050 bytes

2. Tunnel Mode

Tunnel mode encapsulates the entire original packet and adds 50 bytes of overhead.

$1020+50=1070$ bytes Tunnel packet size=1070 bytes

B. Bandwidth Usage Per Second

Transport Mode

$1050\times20{,}000=21{,}000{,}000$ bytes/s

Convert to bits:

$21{,}000{,}000\times8=168{,}000{,}000$ bits/s 168 Mbps

Tunnel Mode

$1070\times20{,}000=21{,}400{,}000$ bytes/s

Convert to bits:

$21{,}400{,}000\times8=171{,}200{,}000$ bits/s 171.2 Mbps

Comparison

$171.2-168=3.2$ Mbps

Tunnel mode consumes **3.2 Mbps more bandwidth**.

C. Efficiency and Security Comparison

Feature	Transport Mode	Tunnel Mode

Feature	Transport Mode	Tunnel Mode
Original packet	1020 bytes	1020 bytes
IPSec overhead	30 bytes	50 bytes
Final size	**1050 bytes**	**1070 bytes**
Bandwidth	**168 Mbps**	**171.2 Mbps**
Efficiency	Higher	Slightly lower
Original IP header	Remains visible	Encapsulated/protected
Security/privacy	Good	Higher
Typical use	Host-to-host	VPN/network-to-network

Transport Mode Efficiency

Overhead percentage:

102030 $\times 100 = 2.94\%$

Therefore, approximately **97.06%** of the original packet size is retained as useful original packet content.

Tunnel Mode Efficiency

102050 $\times 100 = 4.90\%$

Therefore, approximately **95.10%** of the original packet size is retained relative to the original packet.

Thus, **transport mode is more bandwidth-efficient**.

Security

Transport mode:

- Protects the IP payload.
- Original IP addresses remain visible.
- Suitable when endpoints themselves need to communicate securely.

Tunnel mode:

- Encapsulates the entire original IP packet.
- The original IP header is protected inside the tunnel.
- Provides better privacy and is commonly used for VPNs and communication across untrusted networks.

D. Recommendation

For Internal Communication → Transport Mode

For trusted internal networks, **Transport Mode is recommended** because:

- Lower overhead
- Lower bandwidth consumption
- Better efficiency
- Suitable for host-to-host IPSec communication

Here, it uses:

168 Mbps

instead of 171.2 Mbps.

For External Communication → Tunnel Mode

For communication across the Internet or other untrusted networks, **Tunnel Mode is recommended** because:

- The complete original packet is encapsulated.
- Original addressing information is protected from external observers.
- It is well suited to **site-to-site and remote-access VPNs**.
- Provides stronger privacy across untrusted networks.

It uses:

171.2 Mbps

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4. S/MIME Certificate Management in Large Organization
An organization manages 12,000 employees using S/MIME certificates. Each certificate is 2 KB, and certificate renewal happens every year. During renewal, both old and new certificates are stored simultaneously for 1 month.

Task:
A. Calculate total storage required during the renewal period.
B. Estimate certificate distribution overhead across the network.
C. Analyze challenges in certificate lifecycle management at scale.
D. Propose optimization strategies (e.g., automation, centralized PKI, or short-lived certificates).

ANSWER:

Given

- Number of employees = **12,000**
- Certificate size = **2 KB**
- Renewal period = **1 month**
- During renewal, **old + new certificates** are stored simultaneously.
- Renewal occurs once per year.

Assume:

1 KB=1024 bytes

A. Total Storage Required During Renewal

During the renewal period, each employee has **2 certificates**:

2×2=4 KB/employee

For 12,000 employees:

12,000×4=48,000 KB

Convert to MB:

102448,000 =46.875 MB

Therefore:

46.875 MB

Normal Storage

Outside the renewal period, only one certificate is stored:

12,000×2=24,000 KB =23.4375 MB

Thus, during renewal, storage approximately **doubles**:

46.875 MB

B. Certificate Distribution Overhead

Each new certificate is distributed once to each employee:

12,000×2 KB=24,000 KB =23.4375 MB

So, distributing all **12,000 new certificates** requires approximately:

23.44 MB

of network data, ignoring protocol headers, encryption overhead, retransmissions, and other network traffic.

If old certificates are also redistributed

If both old and new certificates were transmitted:

$12{,}000 \times 4\text{ KB} = 48{,}000\text{ KB} = 46.875\text{ MB}$

However, normally only the **new certificates** need to be distributed during renewal.

C. Certificate Lifecycle Management Challenges

Managing S/MIME certificates for 12,000 employees creates several challenges.

1. Certificate Expiration

Certificates expire periodically. Missing a renewal can prevent users from sending or receiving properly signed/encrypted email.

2. Large-Scale Renewal

Renewing thousands of certificates manually is time-consuming and can result in configuration errors.

3. Certificate Revocation

Certificates must be revoked when:

- An employee leaves the organization.
- A private key is compromised.
- A device is lost.
- An employee changes roles.

4. Key Management

Private keys must remain protected. Losing a private key can make encrypted historical email difficult or impossible to decrypt.

5. Distribution

New certificates must be securely distributed to employee devices and email systems.

6. Multiple Devices

Employees may use laptops, phones, tablets, and webmail, creating additional certificate deployment and synchronization requirements.

7. Compatibility

Different email clients and operating systems may handle S/MIME certificates differently.

D. Optimization Strategies

1. Automated Certificate Renewal

Use automated certificate enrollment and renewal rather than manually renewing certificates.

Certificate Expiration
↓
Automatic Detection
↓
New Certificate Generation
↓
Automatic Deployment
↓
Old Certificate Retired

This reduces human errors and administrative effort.

2. Centralized PKI

A centralized **Public Key Infrastructure (PKI)** can manage:

- Certificate generation
- Certificate distribution
- Renewal
- Revocation
- Certificate policies
- Monitoring

This is highly suitable for an organization with **12,000 employees**.

3. Short-Lived Certificates

Short-lived certificates reduce the time window in which a compromised certificate can be abused.

However, they require **highly automated renewal**, otherwise frequent expiration could increase administrative burden.

4. Certificate Revocation Management

Use mechanisms such as:

- **CRL (Certificate Revocation List)**
- **OCSP (Online Certificate Status Protocol)**

to check whether certificates are still valid.

5. Centralized Certificate Inventory

Maintain a database containing:

- Employee identity
- Certificate ID
- Issue date
- Expiration date
- Revocation status
- Device information

This makes it easier to identify certificates that require renewal.

6. Secure Key Backup

Enterprise policies should securely manage private keys, especially when encrypted email must remain accessible after certificate renewal.

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5. Email Encryption Latency in Cloud-Based Systems
A cloud email service encrypts outgoing emails using AES (data) and RSA (key exchange). AES encryption takes 0.2 ms per MB, and RSA key exchange takes 5 ms per email. Each email has 8 MB attachment. The system processes 8,000 emails per hour.

Task:
A. Calculate total encryption time per email.
B. Estimate total processing time per hour.
C. Analyze the performance bottleneck between symmetric and asymmetric operations.
D. Recommend optimization strategies (e.g., ECC adoption, parallel processing, or hardware acceleration).

ANSWER:

Given

- Attachment size = **8 MB/email**
- AES encryption time = **0.2 ms/MB**
- RSA key-exchange time = **5 ms/email**
- Emails processed = **8,000 emails/hour**

A. Total Encryption Time per Email

AES Encryption Time

8 MB×0.2 ms/MB =1.6 ms

RSA Key Exchange Time

=5 ms

Total Time

1.6+5=6.6 ms

Therefore:

6.6 ms/email

B. Total Processing Time per Hour

The system processes **8,000 emails/hour**.

8,000×6.6=52,800 ms

Convert to seconds:

100052,800 =52.8 seconds

Therefore:

52.8 seconds/hour

So, under the assumption that processing is sequential, encryption operations consume **52.8 seconds of compute time per hour**.

Breakdown

AES:

$8,000 \times 1.6 = 12,800 \text{ ms} = 12.8 \text{ seconds/hour}$

RSA:

$8,000 \times 5 = 40,000 \text{ ms} = 40 \text{ seconds/hour}$

Total:

$12.8 + 40 = 52.8 \text{ seconds/hour}$

C. Performance Bottleneck Analysis

AES

AES takes:

1.6 ms/email

Percentage of total:

$\frac{1.6}{6.6} \times 100 \approx 24.24\%$

RSA

RSA takes:

5 ms/email

Percentage:

$\frac{5}{6.6} \times 100 \approx 75.76\%$

Operation	Time/email	Percentage
AES encryption	**1.6 ms**	**24.24%**
RSA key exchange	**5 ms**	**75.76%**
Total	**6.6 ms**	**100%**

Bottleneck

RSA is the main performance bottleneck.

Although the 8 MB attachment requires processing a relatively large amount of data, AES is highly efficient. RSA is computationally more expensive and consumes approximately **76% of the encryption time** in this scenario.

D. Optimization Strategies

1. Adopt ECC

Replace RSA-based key exchange with **Elliptic Curve Cryptography (ECC)** where appropriate.

ECC can provide strong security with smaller keys and can reduce computational and communication overhead compared with traditional RSA configurations.

Thus:

RSA Key Exchange
↓
Replace with
↓
ECC-based Key Exchange
↓
Lower asymmetric overhead

2. Parallel Processing

Instead of processing emails sequentially, multiple emails can be encrypted simultaneously.

For example:

8000 Emails
↓
| CPU 1 | CPU 2 | CPU 3 | CPU 4 |
↓
Parallel Encryption

This reduces **wall-clock processing time**, although the total computational work remains approximately the same.

3. Hardware Acceleration

Use CPUs or cryptographic accelerators that support hardware-assisted AES and public-key cryptography.

This can reduce:

- Encryption latency
- CPU utilization
- Server response time

4. Reuse Secure Sessions/Keys Where Appropriate

Avoid unnecessary asymmetric operations for every email when the protocol and security policy allow secure session or key reuse. The expensive public-key operation should primarily be used to establish or protect symmetric keys.

5. Cloud Auto-Scaling

For a cloud email service, additional encryption workers can be dynamically added when email traffic increases.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context

Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation